



## ParkingSystem

```
- activeVehicles : list <Vehicle>
- slots : vector <ParkingSlot>
- entryQueue : queue <Vehicle>
- recentExitStack : stack <String>
- activeTickets : unordered_map
                        <String, Ticket>
+ lastTicket : String
+ ParkingSystem()
+ addVehicleToQueue(v: Vehicle): void
+ processEntry(): void
+ processExit(...): void
+ showRecentExit(): void
```

## ParkingSlot

```
- slotId : int
- size : int SlotSize
- isOccupied : bool
+ ParkingSlot(id: int, s: SlotSize)
+ getId(): int
+ getSize(): SlotSize
+ getOccupied(): bool
+ setOccupied(status: bool): void
```

Composite

1-\*

## Vehicle

Aggregate

1-\*

```
+ plateNumber : string
+ requiredSize : SlotSize
+ Vehicle(plate: String,
           size: SlotSize)
```

1-1

1-Association

1-1

